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DESIGN AND FABRICATION OF A MICRO-KNIFE FOR MINIMALLY INVASIVE SURGERY

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ABSTRACT

In this paper, a micro-knife for minimally invasive surgery is proposed. It is a silicon horn shape and works in an ultrasonic longitudinal vibration mode to realize the precise incision with its sharp tip at one end. Finite element method in ANSYS is used to analyze and design the device to obtain the vibration frequency and displacement amplitude of the longitudinal mode. In the end, microfabrication of the proposed micro-knife is presented.

1. INTRODUCTION

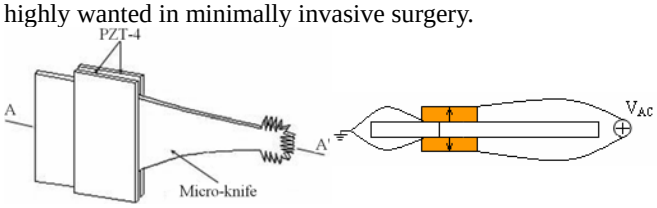
Minimally invasive surgery [1] is a surgical procedure and technique that minimizes the trauma of healthy tissue during the operation. It is in growing demand due to its advantages over the traditional surgical technology in some special applications. It can avoid the wound from widening and bleeding under some levels of pressure against the tissue. Compared with other methods of minimally invasive surgery, such as electrosurgery and laser surgery, ultrasonic surgery works under low temperature and minimizes the heat damage to healthy tissues. Furthermore, it does not stick to the tissue during operation because of the constant motion during the cutting, and has homeostatic effect.

Recently there has been an increased interest in the use of MEMS for surgery due to the rapid development of MEMS technology during the past 20 years. MEMS technology has the potential to not only improve the functionality of existing surgical tools, but also add new capabilities. Some microsurgerical tools have already been developed, such as micro active forceps for eye microsurgery [2], a multifunction microscale system based on silicon micromachining [3], and a MEMS tactile sensor for surgery instrument [4]. More and more research groups are joining this effort.

In this paper, we propose a new micro-knife for minimally invasive surgery. It is a silicon horn shape and adhered with a pair of bulk PZT plates. As an ultrasonic microsurgery tool, micro-knife is based on silicon-micromachined high-intensity actuators [5] that can produce high intensity ultrasound due to material properties of silicon, such as high yield strain, low internal losses, and high speed of sound. When it is driven into large vibrational amplitudes, silicon will not get as hot as some kinds of metals, such as titanium alloys, due to low acoustic internal loss of silicon, furthermore, its sharp tip at one end will move back and forth for precise incision that can be used in minimally invasive surgery. We will investigate its design and microfabrication in the paper.

2. DESIGN OF THE MICRO-KNIFE

In our study, ultrasonic horn structure [6] is selected as the basic structure for micro-knife. As shown in Figure 1, the knife is designed and fabricated into a silicon horn shape, with its zigzag or triangle tip as cutting blade, and two bulk PZT-4 ceramic plates are adhered to the displacement nodal point of the micromachined silicon substrate to excite the device into vibration; by setting up the piezoelectric plates with their polarization in opposite directions, the moments generated by each plate are cancelled while the extensions are added. When a sinusoidal AC signal with appropriate frequency and amplitude is applied onto the PZT-4 plates, the bulk mode of the entire microdevice can be excited and the silicon horn vibrates longitudinally due to the d_{31} coupling from the bounded PZT-4 ceramic plates. Thus, its sharp tip will vibrate, moving back and forth longitudinally to penetrate and cut skin muscle or other biological tissues in the microscale, which is highly wanted in minimally invasive surgery.



(a) (b)

Figure 1. Schematic diagram of the proposed new micro-knife for minimally invasive surgery: (a) 3-D graph; (b) A-A' cross-section.

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As shown in Figure 2, the shape of the microsurgical knife can have four choices, including the symmetrical and unsymmetrical exponential shapes with a zigzag tip, and the symmetrical linear shape with a triangle tip. According to Eisner [6], when the horn is driven into longitudianl vibration the ultrasonic vibration energy can be concentrated on its tip and thus the tip can achieve the maximum displacement. In our study, these four shapes have the same length and width, which are 20mm and 8mm respectively. All four types of micro knife can be fabricated on a 4-inch silicon wafer, thus their thicknesses are all 450μm. At the stub end of the silicon horn structure, the PZT-4 plates can be attached. The length of the PZT-4 plate is 6mm, the width is 10mm, and the thickness is 0.5mm. For the exponential shape, the lengthes of the silicon stub and the horn are 5mm and 13.5mm, respectively, and tip of the horn is 1.5mm in width. For the zigzag micro-blazes, the base of each tooth is 0.3mm and the height is 1.5mm. For the linear shape, the lengthes of the silicon stub and horn are 5mm and 15mm, respectively, and its tip is 50nm in width.

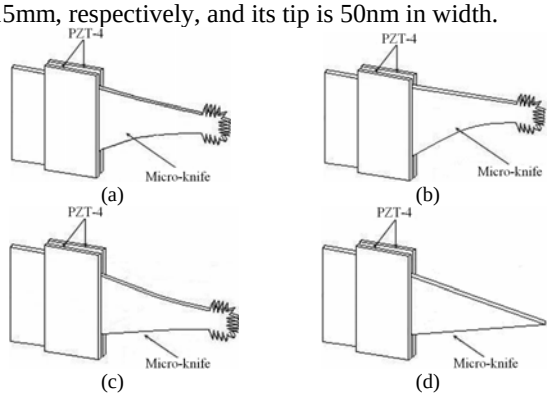

Figure 2. Different shapes of the proposed micro-knife for minimally invasive surgery: (a) partially unsymmetrical exponential shape with a zigzag tip; (b) totally unsymmetrical exponential shape with a zigzag tip; (c) symmetrical exponential shape with a zigzag tip; (d) symmetrical linear shape with a triangle tip.

We use FEM in ANSYS to study the fundamental longitudinal vibration mode of the proposed four structures. In ANSYS, the element type Solid 5 is used to mesh the PZT plates and Solid 45 to mesh the silicon substrate. A FEM meshed model is shown in Figure 3. The modal analysis in ANSYS is used to calculate the resonance frequency and mode shape of the microfabricated silicon horn, and harmonic response analysis in ANSYS is used to obtain the tip displacement of the micro-knife, i.e. the silicon horn adhered with two PZT-4 plates. A sinusoidal AC signal with a voltage of $100V_{p\text{-}p}$ is applied to PZT-4 plates and swept in a frequency range. If the frequency of the applied AC signal accords with the frequency of the fundamental longitudinal mode of the silicon horn, the tip can get the maximum displacement.

Figure 4 shows the mode shape, resonance frequency and tip displacement in longitudinal direction obtained by using ANSYS for four proposed structures. We can see that all four structures will resonate longitudinally at 180KHz-250KHz, an ultrasound range, and their tip displacements are 2-4μm, which are all in the micrometer scale. As expected, the tip displacement of the symmetrical linear shape is the biggest since its tip width is 50nm, the smallest one among these four structures. The tip displacements for other three exponential shapes are slightly smaller than the linear shape. It is noticed that tip widths of exponential shapes are about 1.5mm, much wider than that of the linear shape. Therefore, it can be expected that the micro-knife with an exponential shape can bear much larger stress when used for cutting.

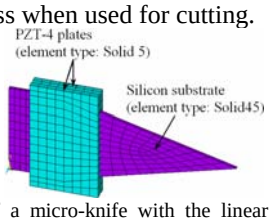

Figure 3. FEM model of a micro-knife with the linear shape silicon horn structure

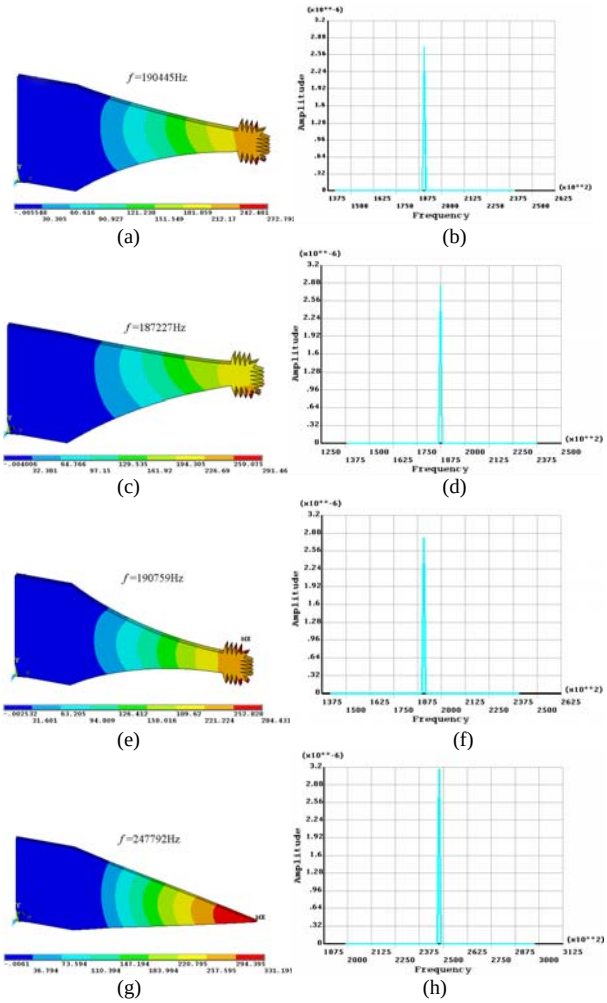

Figure 4. The vibration mode shape, resonance frequency and tip displacement amplitude in longitudinal direction of different shapes of the micro-knife: (a) and (b) show the mode shape and tip displacement amplitude of the partially unsymmetrical exponential shape with a zigzag tip; (c) and (d) show the mode shape and tip displacement of totally unsymmetrical exponential shape with a zigzag tip; (e) and (f) show the made shape and tip dispalcement amplitude of the symmetrical exponential shape with a zigzag tip; (g) and (h) show the mode shape and tip displacement amplitude of the symmetrical linear shape with a triangle tip

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To better compare these four types of micro knife structures, we list the results of FEM analysis in Table 1. We can see that when the shape of the knife is symmetrical there is no any displacement in transverse direction, but if the shape of the knife is unsymmetrical, the displacement in transverse direction will occur besides the longitudinal direction, which is about 0.3μm, less than one third of the displacement amplitude in longitudinal direction. This suggests that the micro-knife in a symmetrical shape may be more suitable for penetration and the micro-knife in an unsymmetrical shape more suitable for cutting.

Table 1 Performance of micro knives with different horn shapes

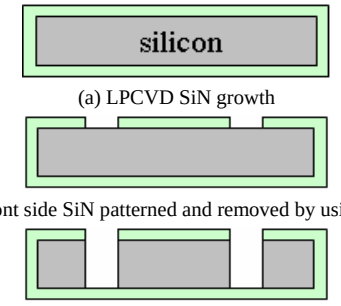
Horn type	Resonance frequency (kHz)	Longitudinal amplitude of tip (μm)	Transverse amplitude of tip (μm)
Partially unsymmetrical Exponential	190.4	2.77	0.29
Totally unsymmetrical exponential	187.2	2.84	0.94
Symmetrical exponential	190.8	2.80	0
Symmetrical linear	247.8	3.16	0

3. DEVICE FABRICATION

The proposed micro-knife is fabricated by using silicon bulk micromachining technology. The fabrication processing is shown in Figure 5. The silicon nitride thin film of 1.3μm is first deposited on the silicon substrate by using LPCVD method, and then the backside of the wafer is patterned and silicon nitride is removed by using RIE (reactive ion etching). After that, deep-RIE is used to etch the bulk silicon until the horn structures are suspended by thin silicon nitride membrane on the wafer. The silicon knives can be obtained by gently breaking the silicon nitride membrane. Figure 6(a) shows a wafer with many fabricated microdevices, and Figure 6(b) shows the micrograph of part of a zigzag tip of a fabricated silicon knife. Once the PZT-4 plates are bonded to the micromachined silicon horn, we can get the micro knife that we designed. Compared with other ultrasonic mcrodevices, the fabrication processing of the proposed micro knife is compatible to CMOS process. Thus, this micro knife has the potential to integrate the related microsensors and circuits on one chip to own more functions.

4. CONCLUSIONS

In this paper, a new silicon-based micro knife for minimally invasive surgery is proposed. Its structure is based on silicon horn shape, and we use FEM in ANSYS to analyze four silicon horn shapes for the design of the proposed micro knife. The results show that this new micro knife can work in the fundamental longitudinal vibration mode, with its microfabricated tip to resonate in the micrometer scale that can be used to penetrate or cut the skin or other biological tissues for minimally invasive surgery.

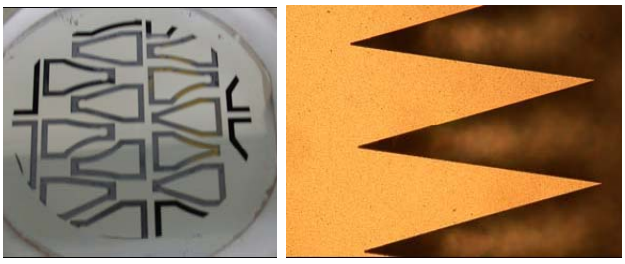


(a) LPCVD SiN growth

(b) front side SiN patterned and removed by using RIE

(c)bulk silicon etched by using deep-RIE

Figure 5. Fabrication processes of the proposed micro-knife



(a) (b)

Figure 6. The microfabricated device: (a) the fabricated silicon micro knives on a wafer; (b) micrograph of part of a microfabricated zigzag tip.

ACKNOWLEDGEMENTS

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